



Reverse Logistics System for Personal Assistance Robots

Case Study Overview

You have been tasked by the CEO of a cutting-edge robotics company to design a comprehensive reverse logistics system for a new family of personal assistance robots aimed at the elderly population. These robots are designed to enhance the quality of life for older individuals, enabling them to live independently at home or in assisted living facilities. The company sells and leases robots and seeks a reverse logistics system that balances customer service, sustainability, cost, asset recovery, and long-term competitive advantage. Production to start in 2029. Basic information:

- **Target Market:** Individuals aged 70 and above, with an expected adoption rate of 2-7%.
- **Product Range:**
 - Basic Model: \$50,000
 - Advanced Model: \$100,000
- **Expected Product Life:** 12 years
- **Distribution/Sales Channels:** Omnichannel: Online sales with delivery via UPS; retail stores in regions with a high percentage of affluent elderly populations.
- **Robotic Features:** The robots are equipped with a communication system that reports their condition and maintenance needs to the company, enabling proactive customer support and lifecycle management.



<https://australiancarersguide.com.au/robotics-in-aged-care/>

Task Objectives

Your job is to design a reverse logistics system that efficiently manages three key scenarios:

1. **Repair and Maintenance:** Handling warranty repairs and paid services for malfunctioning robots.
2. **Leasing Return:** Managing robots that are returned at the end of their lease period, ensuring they are retrofitted for future use.
3. **End-of-Life Management:** Dismantling robots for parts and recycling materials when they reach the end of their operational life.



While the company plans a global presence, your proposal should only consider the US.
Feel free to conceptualize the robot in any fashion you prefer.



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Supply Chain Management

Strategic Constraints

Your network must satisfy the following constraints: Maximum initial reverse-logistics infrastructure investment budget of \$250 million; repair turnaround time of 72 hours or less; lease-return processing within 14 days; material recovery target of at least 90% for end-of-life units; nationwide coverage within the continental United States.

Deliverables

Your submission should include the following components:

1. Demand/customer/output/product assumptions
 - Describe specific assumptions made about the overall demand, the return rates, and the actual lease durations. Describe production, return, and other rates based on these assumptions over a reasonable time horizon (up to 2050?). Include information about geographical distribution of customers, sales versus leases, end of life recovery volumes.
2. Description of Main Steps in the Reverse Logistics Process:
 - Outline the key stages involved in the reverse logistics system from the point of return to final disposition.
 - Include at least two flowchart/figures that graphically describes the proposed macro level steps.
3. Proposal for the Reverse Logistics Network:
 - Facilities: Designate locations for repair centers, refurbishment facilities, and recycling plants. Recommend no more than six company-owned facilities. Justify locations using demographics, logistics infrastructure, labor, cost and service considerations
 - Material Handling: Discuss the processes for receiving, inspecting, and processing returned robots. Include considerations for safety and efficiency. Include at least one self-developed flowchart of these processes.
 - Transportation: Propose transportation methods for returning robots to facilities. Evaluate parcel carriers, dedicated fleets, third-party logistics providers and collection partners. Explain transportation tradeoffs between cost and responsiveness.
4. Role of IT:
 - Explain how the communication system with the robots can aid in forecasting maintenance needs and predicting end-of-life scenarios.
 - Discuss the role of data analytics and AI in improving the efficiency of the reverse logistics process.
5. Challenges and Solutions:
 - Identify potential challenges in implementing the reverse logistics system, such as customer compliance, logistics costs, and inventory management.
 - Propose solutions to mitigate these challenges, including incentives for customers to return products and partnerships with local service providers.
6. Conclusions:
 - Summarize the key elements of your RL system design.